

APPLICATIONS

A COMPUTER-BASED METHOD USING THE BOX-COX TRANSFORMATION FOR IDENTIFYING OUTLIERS

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Summary: The Box-Cox transformation is studied with the objective of identifying outliers in a data set. It is important to determine whether the evidence for the particular transformation is spread evenly throughout the data or is limited to a few cases only. In addition to approximate measures for identifying outliers in literature, computation of the exact likelihood distance is made possible by the powerful processing capability of modern computers. The feasible solution algorithm (FSA) for least trimmed squares regression of Hawkins (1994) is adapted for Box-Cox transformed data to identify more than one outlier. The method is illustrated by applying it to real data.

1. Introduction

When a linear model is fitted to data, important assumptions are usually those of normality and homoscedasticity of the response variable (Y). One way of trying to fulfil these assumptions is to transform Y using the so-called Box-Cox transformation. However, influential cases (or outliers) can firstly cause Y to be non-normal and heteroscedastic on the original scale and, can

secondly give rise to an incorrect transformation when performed with such cases included. Another danger with outliers is that they can give rise to the fitting of a wrong model. Cook and Wang (1983) gave an artificial example to illustrate this, which we will also report below.

Let Y be the response vector with n components, X being the $n \times (k+1)$ design matrix with first column consisting of ones and the $(j+1)$ -th column the values of x_j , while β is the $(k+1)$ -vector of regression coefficients. With ε , the n -vector of random errors, multivariate normally distributed with zero mean vector and covariance matrix $\sigma^2 I$, I the identity matrix, the general linear model is given by

$$Y = X\beta + \varepsilon. \quad (1)$$

In model (1) Y now has an n -variate normal distribution. If the Y_i 's are not normally distributed, let us assume that it is possible to determine a value of λ such that the transformed variable

$$h_\lambda(Y_i) = \begin{cases} \frac{Y_i^\lambda - 1}{\lambda \dot{Y}^{\lambda-1}} & \text{if } \lambda \neq 0 \\ \log(Y_i) \dot{Y} & \text{if } \lambda = 0 \end{cases}, \quad (2)$$

has a normal distribution (Box & Cox, 1964, 1982). Here $\dot{Y}$ is the geometric mean of the Y_i 's, used to normalise the transformed variable to be scale invariant.

With this new model (Box & Cox, 1964 and Cook & Wang, 1983)

$$h_\lambda(Y) = X\beta_\lambda + \varepsilon, \quad (3)$$

the maximum likelihood estimators (MLE) for β_λ , σ_λ^2 and λ are:

$$\hat{\beta}_\lambda = (X'X)^{-1} X'h_\lambda(Y). \quad (4)$$

$$\hat{\sigma}_\lambda^2 = \frac{1}{n} \|h_\lambda(Y) - X\hat{\beta}_\lambda\|^2. \quad (5)$$

$$\hat{\lambda} = \underset{\lambda}{\arg \min} (h_\lambda(Y)'(I-H)h_\lambda(Y)), \quad (6)$$

with H , the so-called hat matrix, defined as $H = X(X'X)^{-1}X'$.

From (6) $\hat{\lambda}$ is obtained as that value of λ which minimizes the sum of squares of residuals of the linear model in (3) with $\hat{\beta}_\lambda$ substituted for β .

Cook and Wang (1983) simulated a bivariate dataset of 10 points which follow the model $\log Y_i = x_i + \varepsilon_i$. In these data the Y_i 's were not normally distributed, but with the Box-Cox transformation with $\lambda = 0$, the new values $Z_i = \log Y_i$ are normally distributed. They chose $\sigma^2 = 0.01$ and then added an eleventh outlier point which follows a linear model (i.e. $\lambda = 1$, or no transformation). Figure 1 gives a scatterplot of this data together with a least squares fit of model (3), taking $\lambda = 0$ based on the first 10 points. Also displayed on the graph, is a linear model ($\lambda = 1$) using all 11 points. The question which arises now is: which of the two models is the correct one if one does not have the knowledge of how the data have been constructed?

Clearly, point 11 masks the actual model and one wants to identify such an outlier.

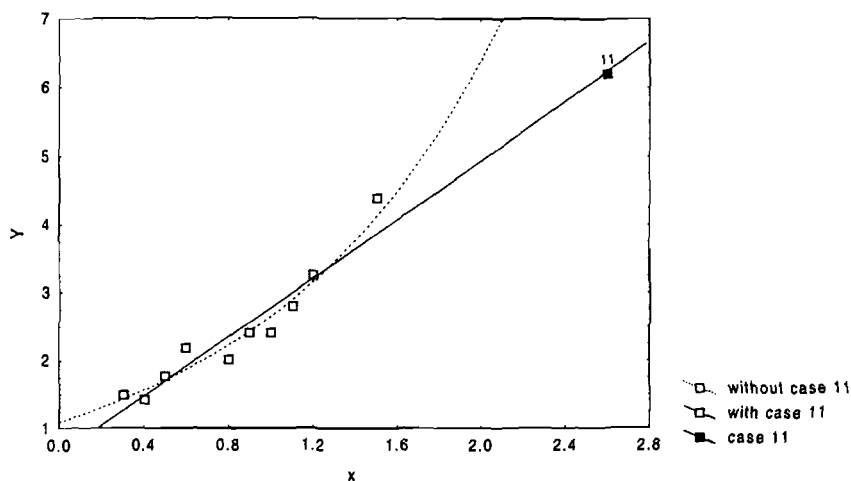


Figure 1. Scatterplot of Cook and Wang's data

In the following section current methods of identifying outliers when using the Box-Cox transformation, are reviewed. Then the feasible solution algorithm (FSA) of Hawkins (1994) is adapted for identification of outliers in the current set-up.

In section 3 properties of the adapted FSA are discussed and applied to some published data sets. It will also be compared to the other methods. Section 4 deals with the practical application of the suggested method, followed by a real-life example.

2. Current methods to identify outliers

Cook and Wang (1993) describe a diagnostic method to measure the influence of cases on the Box-Cox likelihood estimate of the transformation parameter for the response variable in linear regression models. This method is shown to be superior to the diagnostic displays introduced by Atkinson (1982) that do not take into account the leverage structure of the problem.

Hereafter the notation x_I is used to indicate the subvector of any vector x consisting only of the components with indices in $I \subset \{1, 2, \dots, n\}$. On the other hand $x_{(I)}$ indicates that subvector x , of any vector, where the components with indices in I are deleted.

The likelihood distance is used to measure the influence of the deletion of one or more cases in I :

$$\widehat{LD}_{(I)}^{CW} = 2 \left[L_{\max}(\hat{\lambda}) - L_{\max}(\hat{\lambda}_{(I)}^{CW}) \right], \quad (7)$$

where

$$L_{\max}(\hat{\lambda}) = \left[\left\| h_{\hat{\lambda}}(Y) - X\hat{\beta}_{\hat{\lambda}} \right\|^2 / n\hat{Y}^{2(\hat{\lambda}-1)} \right]^{-\frac{n}{2}} (2\pi e)^{-\frac{n}{2}} \quad (8)$$

is the maximum of the likelihood function based on the complete data set. Here $\hat{\lambda}_{(I)}^{(CW)}$, as given by Cook and Wang (1993) is an approximate value of $\hat{\lambda}_{(I)}$, the MLE if one or more observations with indices in I are deleted. The value of $\widehat{LD}_{(I)}^{(CW)}$ will enlarge whenever $\hat{\lambda}_{(I)}^{(CW)}$ deviates from $\hat{\lambda}$ after deletion of the observations with indices in I .

In the derivation of the formula for $\hat{\lambda}_{(I)}^{(CW)}$, the first order Taylor series expansion about $\hat{\lambda}$ is used as an approximation for $h_{\lambda}(Y)$:

$$h_{\lambda}(Y) \cong h_{\hat{\lambda}}(Y) + (\lambda - \hat{\lambda})W_{\hat{\lambda}}(Y)$$

where

$$W_{\lambda_0}(Y_i) = \frac{\partial h_\lambda(Y_i)}{\partial \lambda} \Big|_{\lambda=\lambda_0} = \begin{cases} \frac{Y_i^{\hat{\lambda}} \left[\lambda \log \frac{Y_i}{\dot{Y}} - 1 \right] + \lambda \log \dot{Y} + 1}{\lambda^2 \dot{Y}^{\lambda-1}} & \text{if } \hat{\lambda} \neq 0 \\ \dot{Y} \log Y_i \left[\frac{\log Y_i}{2} - \log \dot{Y} \right] & \text{if } \hat{\lambda} = 0 \end{cases},$$

to obtain

$$\hat{\lambda}_{(I)}^{CW} = \hat{\lambda} + \frac{e'_{h_I} (I - H_I)^{-1} e_{W_I}}{(n - k) s_W^2 - e'_{W_I} (I - H_I)^{-1} e_{W_I}}, \quad (9)$$

where H_I is the submatrix of H corresponding to the rows and columns indexed by I , while e_{h_I} and e_{W_I} are the subvectors of values with indices in I of $e_h = (I - H)h_{\hat{\lambda}}(Y)$ and $e_W = (I - H)W_{\hat{\lambda}}(Y)$, with $s_W^2 = e'_W e_W / (n - k)$.

The approximation for the likelihood distance in (7) is derived from the above expressions to yield

$$\widehat{LD}_{(I)}^{CW} = n \log \left\{ 1 + \frac{e'_W e_W}{e'_h e_h} \left(\hat{\lambda}_{(I)}^{CW} - \hat{\lambda} \right)^2 \right\}. \quad (10)$$

This method was criticized by Atkinson (1987) because it requires the calculation of the MLE $\hat{\lambda}$ by iteration. Today this is no problem due to the improvement in computer technology. The sum of squares of residuals can be computed for all λ -values incremented in a certain interval. As shown in (6), $\hat{\lambda}^E$ is that value of λ for which the error sum of squares is a minimum.

Whenever an approximation of $\hat{\lambda}^E$ is found, the increments can be reduced and the error sum of squares calculated for λ -values in an interval around $\hat{\lambda}^E$ until the desired accuracy is reached. In the same way $\hat{\lambda}_{(I)}^E$ can be calculated as that value of $\lambda_{(I)}$ with the minimum error sum of squares for observations with indices in I deleted.

The likelihood distance $LD_{(I)}^E = 2 \left[L_{\max}(\hat{\lambda}^E) - L_{\max}(\hat{\lambda}_{(I)}^E) \right]$ can be calculated where $\hat{\lambda}^E$ and $\hat{\lambda}_{(I)}^E$ are substituted for $\hat{\lambda}$ in (8).

This gives an easy way to calculate the likelihood distance when one observation is deleted at a time. Whenever two or more observations are deleted simultaneously, the different combinations of deleted values are so many that this method becomes impractical.

Hawkins (1994) developed an algorithm to find that subset of cases for which deletion from the data set would lead to the regression with smallest residual sum of squares. This feasible solution algorithm (FSA) for least trimmed squares regression takes random starting solutions and refines each to the local optimum. Repeating this for different starting sets provides the global optimum.

To adapt the FSA for Box-Cox transformed data, the subset with minimum residual sum of squares of the transformed response variable is found using the following method:

1. Apply the non-normalized Box-Cox transformation

$$Z_i = \begin{cases} \frac{Y_i^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \log(Y_i) & \text{if } \lambda = 0 \end{cases}$$

for a given value of $\lambda = \lambda_0$. Normalization is applied in 4, by using the geometric mean of only the cases included in that specific subset.

2. Choose a random subset J from the data consisting of d cases. Calculate the residual sum of squares for the ordinary least squares (OLS) regression as well as the geometric mean.

3. Exchange (swap) case i from subset J with each case j in $\bar{J}$ (i.e. trim case i and restore case j) and calculate the resulting change in residual sum of squares for each of these swaps, where

$$SSR_{J_{j(i)}} = SSR_J + \Delta SSR_J$$

and

$$\Delta SSR_J = \frac{(1 - h_{ii}) e_{j,j}^2 - (1 + h_{jj}) e_{j,i}^2 + 2h_{ij} e_{j,i} e_{j,i}}{[h_{ij}^2 + (1 + h_{jj})(1 - h_{ii})]}.$$

4. Test if after this swap the residual sum of squares of the normalized response variable

$$SSR_{h_{J_{j(i)}}} = \frac{SSR_{J_{j(i)}}}{\left[\hat{Y}_J \left(\frac{Y_j}{Y_i} \right)^{\frac{1}{d}} \right]^{2(\lambda-1)}}$$

is smaller than

$$SSR_{h_J} = \frac{SSR_J}{\hat{Y}_J^{2(\lambda-1)}}.$$

5. Repeat steps 3 and 4 for each case i in J until there is no possible swap that will reduce the criterion. This subset is called a feasible set.
6. Repeat steps 2 to 5 a large number of times.
7. The feasible solution with smallest residual sum of squares of normalized responses is taken to be the least trimmed squares solution to the Box-Cox transformed data with transformation parameter $\lambda = \lambda_0$.

3. Properties of the methods

3.1 The practical evaluation of the FSA adapted for Box-Cox transformations

Hawkins (1994) has shown that the FSA converges to a global optimum from a large number of starting subsets. In the current case the criterion used is not only dependent on the minimum residual sum of squares, but also on the value of the transformation parameter, λ , so that the investigation for practical performance has to be repeated.

Let π denote the proportion of all starting subsets that will terminate in a specific feasible solution, i.e. the probability to reach a global optimum, while $E(P)$ is the average number of case swops to terminate in that feasible solution when starting with a random subset. It is not possible to calculate these quantities theoretically, therefore all possible subsets of size d are generated and used as the algorithm's starting points. The feasible solution to which it converged as well as the number of case swops required in order to converge, was recorded so that the proportion of all subsets converging to a certain feasible solution and the average number of case swops necessary to reach that feasible solution, can be calculated. The feasible solution with the smallest residual sum of squares provided the least trimmed squares solution.

This method was applied to an expansion of Cook and Wang's data (1983) in which four other outliers were added as cases 12 to 15 and also five other data points generated from the log-transformation with expected value of errors normally distributed with mean 0 and variance 0.01. This data set will be referred to as CW5 and its scatterplot is displayed in figure 2. Another data set that will be used from literature to identify outliers is the salinity data

(SALINITY) consisting of 28 cases where the bi-weekly mean salinity of water (Y) in North Carolina was measured against the following regressors: the salinity of the previous two weeks (x_1), the discharge (x_2) and the volume of water from the river (x_3) (Atkinson, 1987:48).

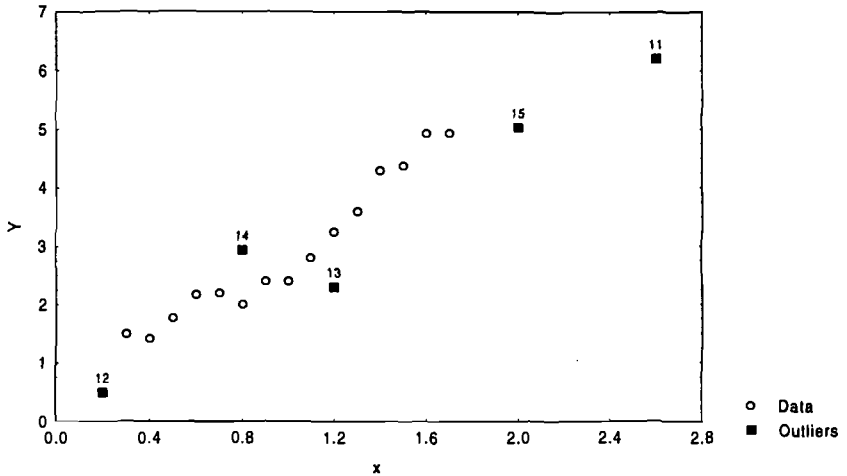


Figure 2. Scatterplot of CW5

This investigation was done for no transformation (i.e. $\lambda = 1$), for a log-transformation ($\lambda = 0$) and for the $\hat{\lambda}_{(I)}$ -values indicated by the exact and approximate Cook and Wang's likelihood distance methods.

From tables I and II it is clear that the probability of reaching the global optimum (π) was large: for both CW5 and SALINITY it was 100% in most of the cases. Hawkins (1994) has shown theoretically that only a few dozen

starting subsets are necessary for convergence to the global optimum when the values of π vary between 4% and 78%. With the high speed computers available today the number of starting subsets is no longer a factor. The low average number of swops, $E(P)$, also indicates how effective the FSA can be.

Table I. Convergence and practical performance of the FSA for the deletion of three cases

(a) CW5

	$\lambda=1$	$\lambda=0$	$\hat{\lambda}_{(I)}^E = -0,071$	$\hat{\lambda}_{(I)}^{CW} = -0,245$
Criterion*	1,635	1,682	1,738	1,916
Count**	1140	1140	1140	1140
$\pi(\%)$	100	100	100	100
$E(P)$	2,852	2,552	2,553	2,550
Cases trimmed	13, 14, 19	11, 12, 13	11, 12, 13	11, 12, 14

(b) SALINITY

	$\lambda=1$	$\lambda=0$	$\hat{\lambda}_{(I)}^E = -0,307$	$\hat{\lambda}_{(I)}^{CW} = -1,1$
Criterion	14,536	15,135	15,395	22,033
Count	3276	3276	3276	3276
$\pi(\%)$	100	100	100	100
$E(P)$	2,721	2,679	2,679	2,677
Cases trimmed	15, 16, 17	3, 15, 16	3, 15, 16	3, 15, 16

* Minimum residual sum of squares.

** Number of starting subsets which terminated to the same solution.

Table II. Convergence and practical performance of the FSA for the deletion of five cases

(a) CW5

	$\lambda=1$			$\lambda=0$	$\hat{\lambda}_{(I)}^E = -0,5$	$\hat{\lambda}_{(I)}^{CW} = -0,89$
Criterion	1,119	1,011	1,148	0,615	0,763	1,148
Count	5965	7636	1903	15504	15504	15504
$\pi(\%)$	38,50	49,3	12,3	100	100	100
E(P)	3,324	3,457	2,749	3,752	3,756	3,766
Cases trimmed	7,11, 13,14, 15	13,14, 18,19, 20	5,7, 12,13, 19	11,12, 13,14, 15	11,12, 13,14, 15	11,12, 13,14, 15

(b) SALINITY

	$\lambda=1$	$\lambda=0$	$\hat{\lambda}_{(I)}^E = -0,63$	$\hat{\lambda}_{(I)}^{CW} = -2,768$	
Criterion	9,266	7,894	10,362	53,318	71,003
Count	98280	98280	98280	94393	3887
$\pi(\%)$	100	100	100	96	4
E(P)	4,148	4,208	4,127	4,238	2,638
Cases trimmed	5,8, 15,16,17	3,13,15 16,17	3,13,15, 16,17	3,4,5, 13,16	3,11,16 23

3.2 Application of the methods to determine one or more outliers

By deleting one data point at a time, $\hat{\lambda}_{(I)}$ was calculated using the Cook and Wang and "exact" methods to determine the influence of individual data points. For CW5 the likelihood distances indicate that cases 11 to 13 are influential with the possibility that cases 1, 7 and 14 can also be outliers (see table III). Three of the real outliers are thus correctly identified. In the SALINITY data case 16

is clearly indicated as influential (see table IV). In most of the cases there is a resemblance between the exact and Cook and Wang's likelihood distances, but in some cases there is a difference. This is an indication that the approximations are in some cases inaccurate and that the exact likelihood distance must be used.

Table III. Comparison of methods for the deletion of one outlier in CW5

I	$\hat{\lambda}_{(I)}^{CW}$	$\hat{\lambda}_{(I)}^E$	$\widehat{LD}_{(I)}^{CW}$	$LD_{(I)}^E$
1	1,061	1,087	0,014	0,063
2	1,033	1,065	0,000	0,021
3	1,034	1,054	0,000	0,009
4	1,021	1,032	0,002	0,000
5	1,054	1,065	0,008	0,021
6	1,053	1,056	0,007	0,010
7	1,083	1,081	0,046	0,050
8	1,071	1,064	0,027	0,020
9	1,043	1,032	0,001	0,000
10	1,024	1,003	0,001	0,020
*11	0,836	0,852	0,722	0,743
*12	0,552	0,743	3,960	1,929
*13	1,173	1,169	0,366	0,386
*14	0,986	0,989	0,041	0,044
*15	1,028	0,998	0,001	0,027
16	1,030	1,040	0,000	0,001
17	1,029	1,015	0,000	0,007
18	1,019	1,001	0,003	0,023
19	1,077	1,046	0,035	0,003
20	1,059	1,027	0,012	0,000

* Constructed outlier cases

Table IV. Comparison of methods for the deletion of one outlier in SALINITY

	$\hat{\lambda}_{(I)}^{CW}$	$\hat{\lambda}_{(I)}^E$	$\widehat{LD}_{(I)}^{CW}$	$LD_{(I)}^E$
1	1,036	1,069	0,018	0,055
2	1,006	1,048	0,005	0,034
3	0,430	0,856	1,023	0,064
4	0,977	1,084	0,000	0,071
5	0,890	1,075	0,020	0,061
6	0,962	1,052	0,000	0,038
7	0,895	0,956	0,018	0,000
8	1,062	1,074	0,033	0,060
9	0,939	0,899	0,002	0,023
10	0,966	0,922	0,000	0,010
11	1,065	1,030	0,036	0,021
12	0,979	0,961	0,000	0,000
13	0,971	0,922	0,000	0,010
14	0,968	0,928	0,000	0,007
15	1,126	1,075	0,093	0,061
16	0,283	0,460	1,645	1,480
17	1,049	1,069	0,025	0,055
18	0,875	0,916	0,029	0,012
19	0,885	0,877	0,023	0,042
20	0,963	0,920	0,000	0,011
21	1,010	0,938	0,007	0,003
22	0,994	0,917	0,002	0,012
23	0,957	0,902	0,000	0,021
24	1,000	0,963	0,004	0,000
25	0,954	0,923	0,000	0,009
26	0,971	0,920	0,000	0,011
27	1,026	0,939	0,013	0,003
28	1,195	1,027	0,190	0,019

When three or five data points are deleted, it is clear from tables V and VI that the approximate likelihood distance indicates other outliers than the

exact likelihood distance. For deletion of three data points in CW5 three of the real outliers are identified with the exact method compared to only two of the real outliers with the approximation. Four of the outliers are correctly identified with the exact method for deletion of five outliers compared to only two correctly identified outliers with the approximate method of Cook and Wang.

The FSA for Box-Cox transformed data when three cases were deleted, identified three of the real outliers in CW5 correctly as outliers for values of the transformation parameter (λ) varying between $-1,0$ and $1,0$. When five outliers are to be identified in CW5 the FSA identifies all five outliers correctly for values of $-1,0 \leq \lambda < 0,8$. For λ -values larger than $0,8$ a few data points are occasionally inaccurately identified as outliers. This shows that the FSA identifies outliers correctly provided that the correct interval within which possible transformation parameters lie, is made. For CW5 and SALINITY it is clear that the FSA identifies a different set of outliers than either the exact or approximate methods.

Table V. Comparison of methods for the deletion of three outliers

	$\hat{\lambda}_{(I)}^{CW}$	$\hat{\lambda}_{(I)}^E$	$\hat{\lambda}_{(I)}^{FSA}$	
CW5	-0,245	-0,071	-0,2	0,4
Cases trimmed	2, 11, 12	11, 12, 15	11, 12, 14	11, 12, 13
SALINITY	-1,100	-0,307	-0,1	0,7
Cases trimmed	3, 4, 5	3, 16, 24	3, 6, 15	15, 16, 17

Table VI. Comparison of methods for the deletion of five outliers

	$\hat{\lambda}_{(I)}^{CW}$	$\hat{\lambda}_{(I)}^E$	$\hat{\lambda}_{(I)}^{FSA}$
CW5	-0,89	-0,50	0,0
Cases trimmed	1, 2, 3, 11, 12	2, 11, 12, 14, 15	11, 12, 13, 14, 15
SALINITY	-2,768	-0,630	0,1
Cases trimmed	3, 4, 5, 6, 17	3, 7, 16, 18, 24	3, 13, 15, 16, 17

4. Practical application of the methods

In practice we suggest that the likelihood distance method should be used to get an idea of the number of outliers in the data. If there is only one outlier in the data, this method identifies it and also gives an indication of the appropriate Box-Cox transformation parameter. When more than one outlier can be identified, we suggest that the number as indicated before should be deleted using the FSA method. In this case the researcher must have an idea of what the appropriate Box-Cox transformation parameter should be, as its value is masked by the outliers in the data. This procedure was applied to the following data:

(1) CW10

This is a further expansion of the Cook and Wang (1983) data in which 25 log-transformed cases were added to CW5 plus another five outliers as cases 46 to 50 added to the existing outliers (no. 11-15). The likelihood distance method identifies cases 11 to 13 as outliers with the slight possibility of another 12 outliers from which cases 14 and 47 to 50 are the constructed outliers. When ten cases are deleted by the FSA method the ten constructed

outliers are deleted for values of $-0,2 \leq \lambda \leq 0,3$ (which include the true value $\lambda = 0$). For smaller or larger λ -values cases that were not real outliers, were occasionally deleted. This indicates the importance of the choice of the transformation parameter. Because 15 possible outliers were identified, the FSA method was used to delete 15 cases. The ten constructed outliers were now between the deleted cases for $-1,0 \leq \lambda \leq 0,4$, again including $\lambda = 0$.

To investigate if the interval of λ -values in which the outliers are correctly identified is broadened if there are fewer outliers in the data, we replaced outlier cases 46 to 50 by log-transformed data with errors $N(0, 0,01)$ distributed. The interval in which cases 11 to 15 were identified as outliers did indeed broaden to $-0,8 \leq \lambda \leq 0,6$, again including $\lambda = 0$, for the deletion of 5 cases. In the same way the interval in which cases 11 and 12 were correctly identified as outliers did broaden to $-2,0 \leq \lambda \leq 0,8$ when only cases 11 and 12, with high leverage, were retained as outliers and 2 cases were deleted. When only cases 14 and 15, with no leverage effect, were retained as outliers the interval in which they were correctly identified as outliers was narrower: $-0,8 \leq \lambda \leq 0,4$. This indicates that this method is very effective for outliers with high leverage, provided that only a small proportion of outliers is present in the data.

(2) Vaal River data

In an environmental study investigating the factors having an influence on the growth of alga in the Vaal River at Balkfontein (Janse van Vuuren, 1996) the MLE for the transformation parameter yields a value of $-0,003$ indicating that a log-transformation may be applicable. A multiple linear

regression of the log of the 109 alga counts on 11 regressors resulted in a regression model with a coefficient of determination (R^2) of 0,383. The likelihood distance method indicated that cases 2, 100 and 106 were outliers.

When three cases were deleted with the FSA method the same three cases were indicated. After deletion of these three cases R^2 improved to 0,472.

5. Conclusions

The adapted FSA method is a quick and easy way to identify outliers especially for more than one outlier in a large data set. From the CW10 data follows that as much as 20% of outliers can be identified correctly for the correct choice of the transformation parameter λ . It is also clear that this procedure is very effective to eliminate outliers with high leverage that can sometimes mask the transformation parameter.

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